Aim: Design a Low pass filter using Hamming and Blackman window.

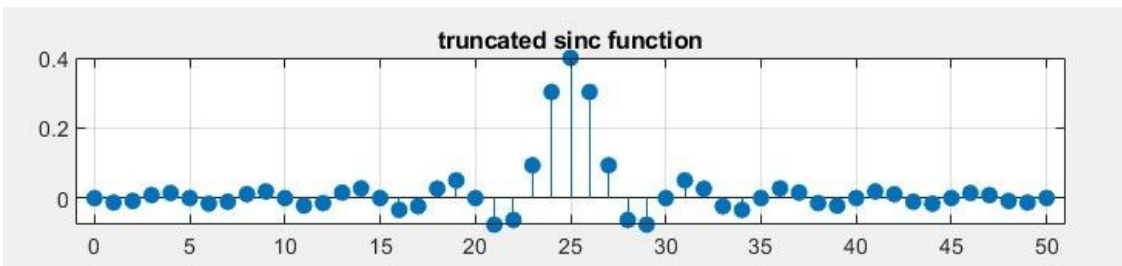
$$z = \frac{\sin(2\pi fcp)}{\pi p}, \text{ where } p = n - \frac{(m-1)}{2}$$

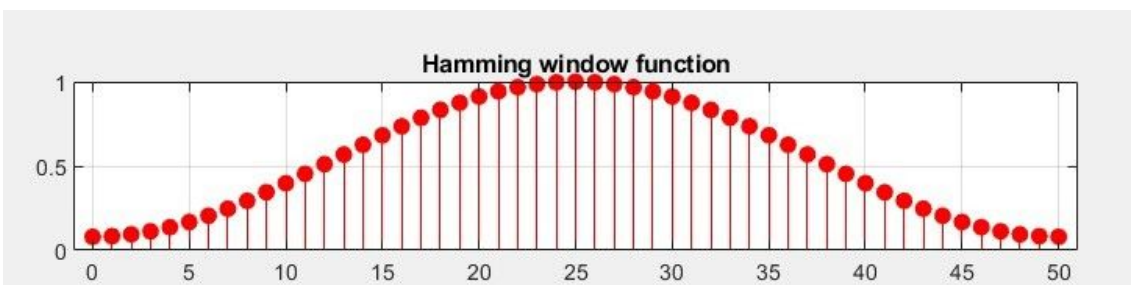
- Consider a truncated sinc function z with angle p as given.
- Plot its truncated unit sample response.
- Plot Frequency response of the truncated sinc function
- Plot the Hamming window function.
- Plot the multiplication of Hamming window and sinc function.
- Frequency response of the windowed sinc function.
- Frequency response of the windowed sinc function in dB.

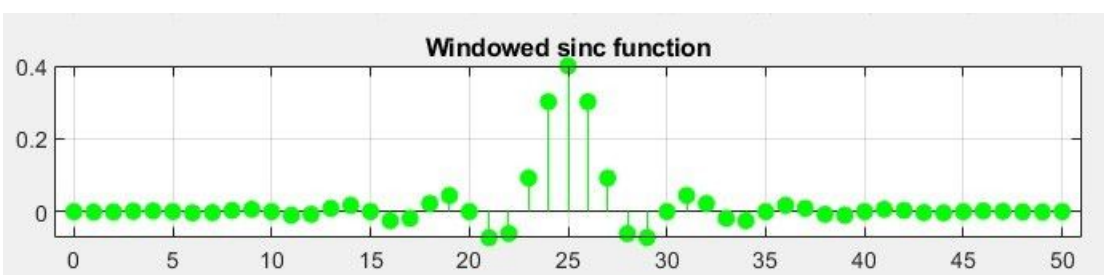
MATLAB CODE AND RESULT :

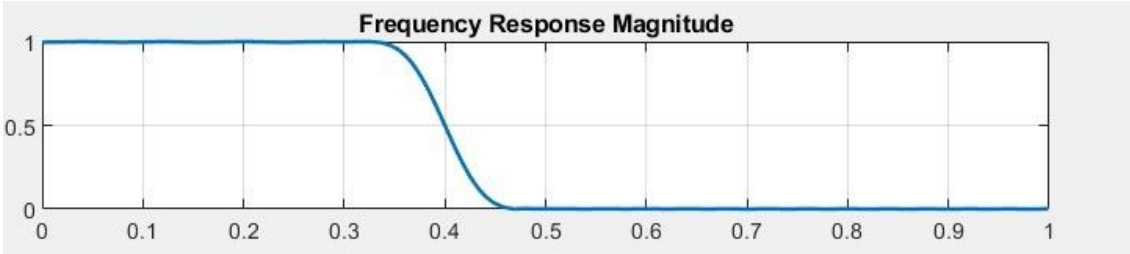
```
m = 51;
fc = 0.2;
n = 0 : m - 1;
p = n - (m - 1)/2;
% Truncated sinc function
z = sin(2*pi*fc*p)./(pi*p); z((m+1)/2)
= 2*fc;


% Windowing function w_hamming
= hamming(m)'; w_blackman =
blackman(m)';

% Windowed impulse response (multiplication)
h_windowed = z.*w_hamming;
h_blackmaned = z.*w_blackman;

% Frequency response
[h_ideal, w] = freqz(z,1,1024); % Original truncated sinc [H_windowed,w] =
freqz(h_windowed, 1, 1024); % Windowed sinc [H_blackmaned,w] =
freqz(h_blackmaned, 1, 1024); % Blackmaned sinc

% Plotting results
% Hamming
Window
subplot(5,2,1);
stem(n,z,"filled"); title("truncated sinc
function"); grid on;
subplot(5,2,2);
stem(n,w_hamming,"filled","r");
title("Hamming window function"); grid on;
subplot(5,2,3);
stem(n,h_windowed,"filled","g");
title("Windowed sinc function"); grid on;
subplot(5,2,4);
plot(w/pi,abs(H_windowed),'LineWidth',1.5);
title("Frequency Response Magnitude");
```

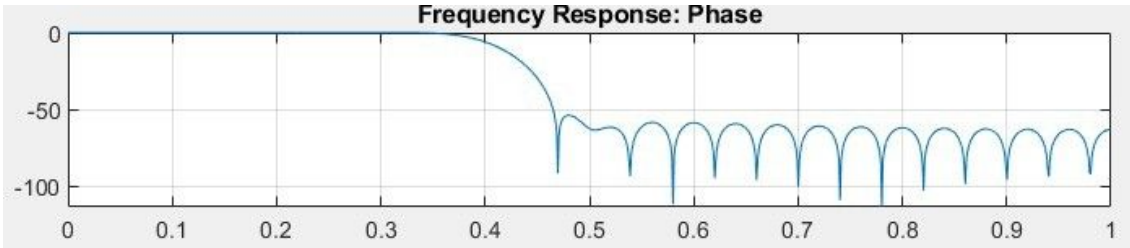
```
grid on;
subplot(5,2,5);
plot(w/pi,20*log10(abs(H_windowed)/max(abs(H_windowed)))); title("Frequency
Response: Phase");
grid on;

subplot(5,2,6);
stem(n,w_blackman,"filled","r");
title("Blackman window function"); grid on;
subplot(5,2,7); stem(n,h_blackmaned,"filled","g");
title("Blackmaned Windowed sinc function"); grid on;
subplot(5,2,8);
plot(w/pi,abs(H_blackmaned),'LineWidth',1.5);
title("Frequency Response Magnitude");
grid on;
subplot(5,2,9);
plot(w/pi,20*log10(abs(H_blackmaned)/max(abs(H_blackmaned)))); title("Frequency
Response: Phase");
grid on;
```

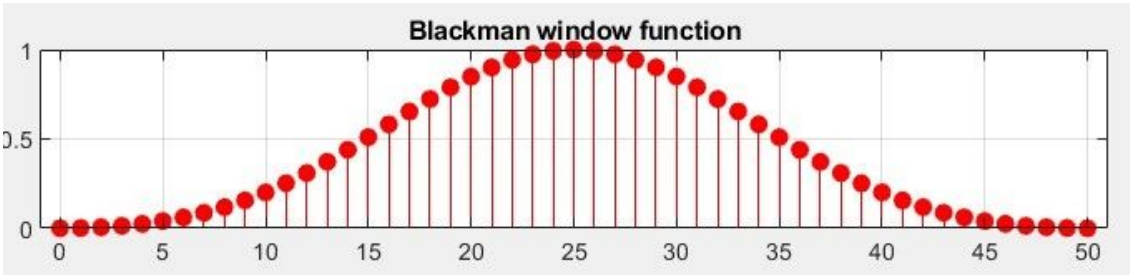


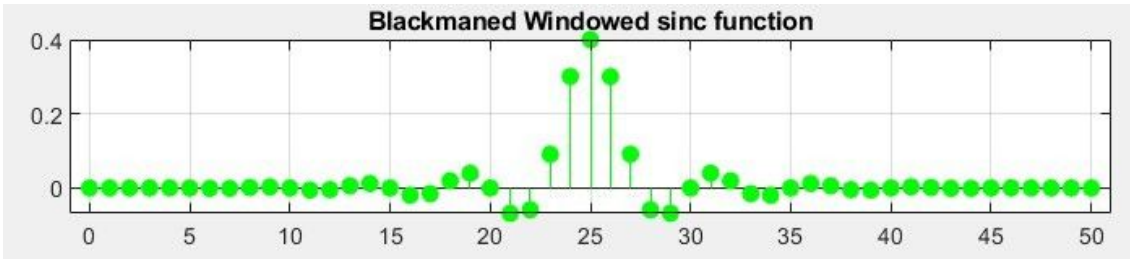


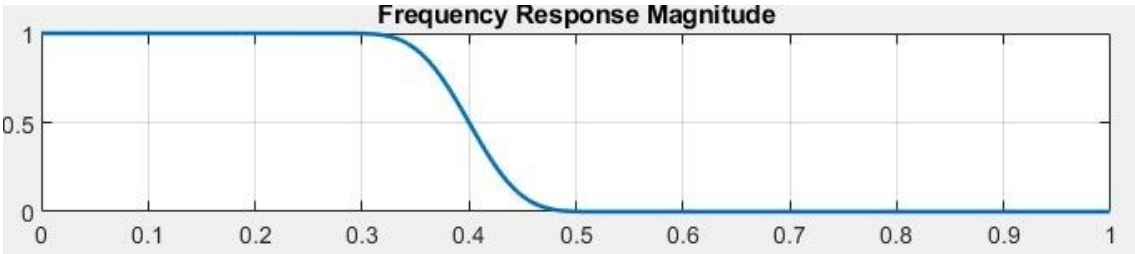


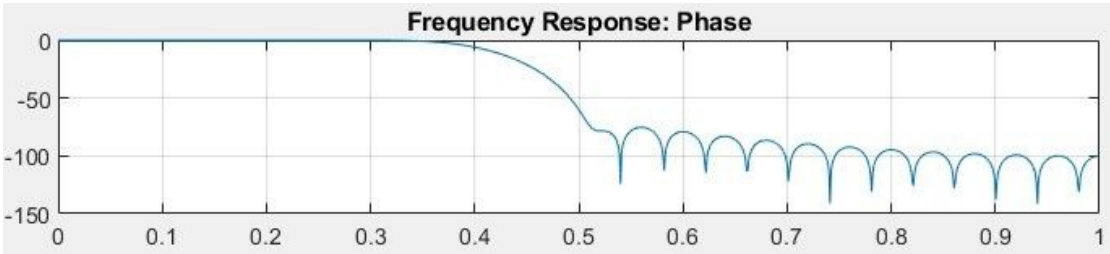

Frequency Response Magnitude
1
0.5
0
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1


Frequency Response: Phase
0
-50
-100
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1


Blackman window function
1
0.5
0
0 5 10 15 20 25 30 35 40 45 50


Blackmaned Windowed sinc function
0.4
0.2
0
0 5 10 15 20 25 30 35 40 45 50


Frequency Response Magnitude
1
0.5
0
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1


Frequency Response: Phase
0
-50
-100
-150
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1

Conclusion : In this experiment, a low-pass FIR filter was designed using the windowing method with Hamming and Blackman windows. The ideal sinc function was truncated and multiplied with the window functions to obtain the filter response. From the frequency response plots, it was observed that the Blackman window provides better stopband attenuation, while the Hamming window has a slightly narrower transition band. Thus, windowing helps control the filter characteristics effectively.